

LINE WORKS AI Adoption Strategy

A Company-Level Analysis for

PX Mart · 全聯福利中心

PRESENTATION

GROUP 1

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LINE WORKS is the enterprise version of LINE



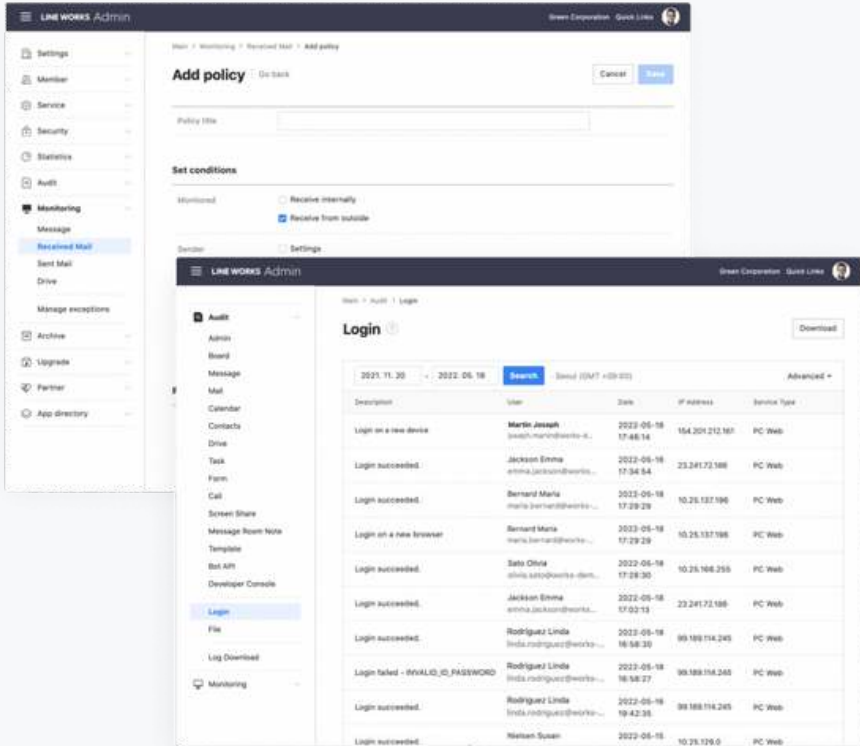
Familiar UX

Same chat paradigm as personal LINE



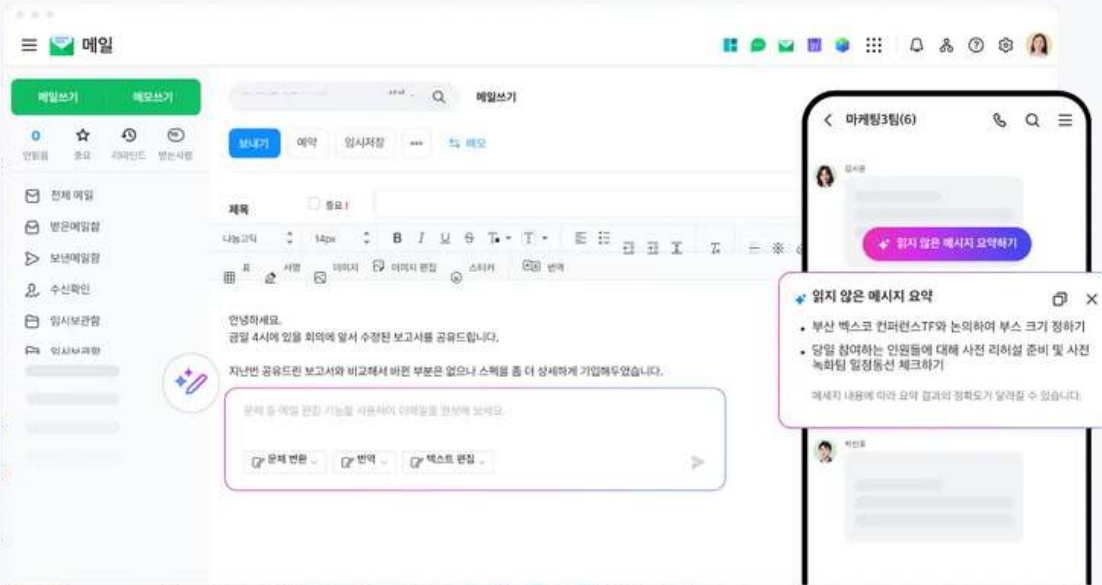
Enterprise core

Admin-grade controls and modules



Native AI

HyperCLOVA-X agents inside chat



Key functions

- chat
- calendar
- drive
- board
- survey
- AI bot
- admin console
- API access
- ISO 27001 / 27017 / 27018

Live in Taiwan

Since October 2025

#1 in Japan

8 consecutive years

5.8M users

Globally · ARR ¥16bn · +40% YoY

590,000

Corporate clients in Japan

Why LINE WORKS is the right platform for PX MART

COMPANIES THAT SHOULD USE LINE WORKS

1

Deskless workforce

Most staff work without a PC

2

Already on LINE

Workforce uses personal LINE for work

3

Distributed multi-site operations

Many sites, fragmented communications

4

Customer-facing operations

Bridge B2C and B2B in one identity

WHY PX MART FITS

~ 50%+ deskless at PX Mart

Part-time workers: approx. 12,117 people (*2024 ESG report*)

People already run on LINE

94% of Taiwan's population uses LINE

1,200+ stores nationwide

Plus 22 RT-Mart hypermarkets to integrate

PX Pay + LINE Official Account already live

Customers already chat with the brand on LINE

PX Mart hits all four criteria, a strong fit for LINE WORKS implementation

PX Mart Overview

BUSINESS DESCRIPTION

- **Largest supermarket chain in Taiwan**
by store count and revenue; 2nd largest retailer overall behind 7-Eleven
- **1,200+ direct-operated stores**
neighborhood-density model across all 22 cities/counties
- **Fresh-food traffic-driver strategy**
Acquired TAPMC to deepen perishables sourcing
- **Omnichannel push via PX Pay**
Proprietary digital wallet captures transaction data and loyalty points

KEY FACTS

Chairman
Lin Ming-Hsiung 林敏雄

Ownership
Private (Yuanlih Group)

Revenue (2024)
NT\$210 billion (US\$6.36 billion)

Workforce
~30,000 (incl. RT-Mart)

KEY MILESTONES

- 1998** Privatized & rebranded as PX Mart under Lin Ming-Hsiung
- 2012** Surpasses Carrefour to become #1 supermarket
- 2018** Launches PX Pay
- 2022** Acquires RT-Mart (NT\$ 11.5 bn) — enters hypermarket
- 2024** Revenue hit a record NT\$210 B

1,200+

STORES NATIONWIDE

Largest scale in Taiwan

- PX Mart operates more stores than any other supermarket chain.
- AI improvement multiplies across 1,200+ locations and millions of transactions.

~64%

SUPERMARKET SHARE of 2021

Dominant and growing

- Largest share of Taiwan's supermarket.
- The NT\$11.5 bn RT-Mart acquisition adds 22 hypermarkets and increases integration.

[Link to the article](#)

~50+%

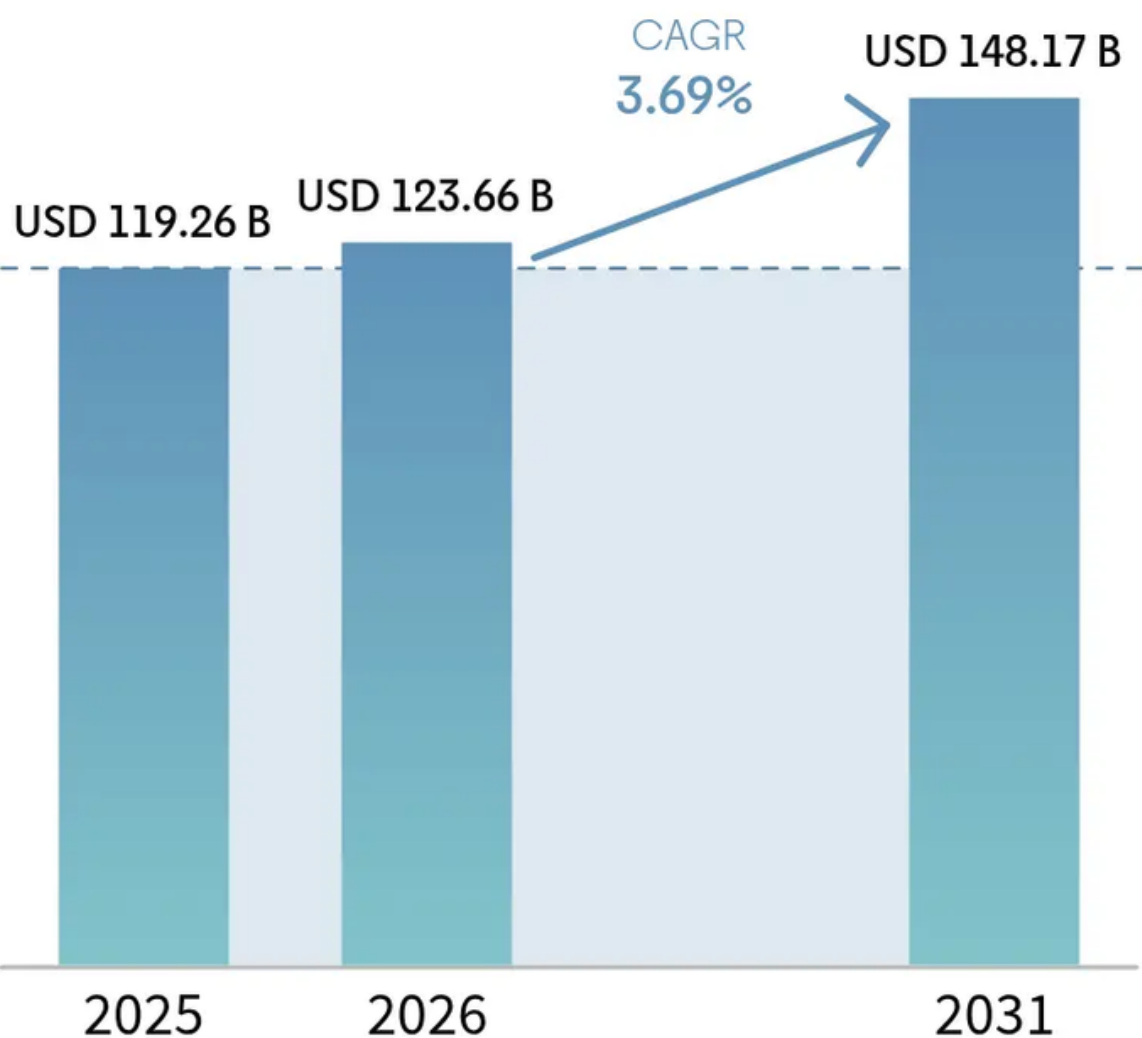
DESKLESS WORKFORCE

LINE WORKS' core target user

- Approximately 60% full-time and 40% part-time staff.
- Store associates handle fresh food, replenishment, cashiering, and customer service on rotating shifts.

Taiwan retail market size in 2026 is estimated at USD 123.66 billion

Taiwan Retail Market
Market Size in USD Billion



Source: Mordor Intelligence



INDUSTRY DYNAMICS

Retail market growth

+3.69% over 2026-2031

Food retail size

USD 38 bn

75% of total is modern market

Market structure

~80% in 2 player

PX Mart + Uni-President control ~40% each post-mergers

E-commerce growth

2.3% in 2025

Grocery now ~17.3% of overall online FMCG sales.

Convenience stores

13,700+

Highest density globally

PX Mart's official priorities × our proposed AI adoption goals

PX MART'S STATED 3-5 YEAR PRIORITIES (cited)

01 Become a 'supermarket for everyone'

Source: PX Mart 2024 annual report

02 Win on fresh food

Source: Chairman Lin interview, Commercial Times 2024

03 Digital transformation via PX Pay

Source: PX Mart press release, 2023

04 Omnichannel integration

Source: PX Mart 2024 annual report

05 Integrate RT-Mart smoothly

Source: FTC filing, 2022

OUR PROPOSED AI ADOPTION GOALS via LINE WORKS

EFFICIENCY GOALS

Cost reduction · time savings · error elimination

- Stop relying on personal LINE for supplier and contact (close security gap)
- Replace manual schedule-building with AI-merged shifts published in LINE WORKS Calendar
- Cut perishable shrinkage 15-20% with AI markdown pricing pushed as live tasks to floor staff

VALUE-CREATION GOALS

New capabilities · differentiation · moats

- HQ-owned LINE WORKS seats by job role
- Faster in-store issue response improves customer experience and brand trust

PX Mart's internal value chain

SUPPORT ACTIVITIES



PRIMARY ACTIVITIES

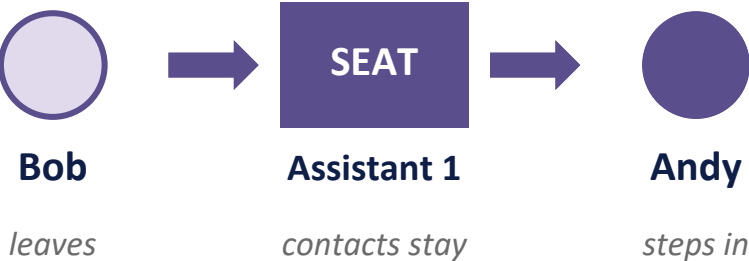
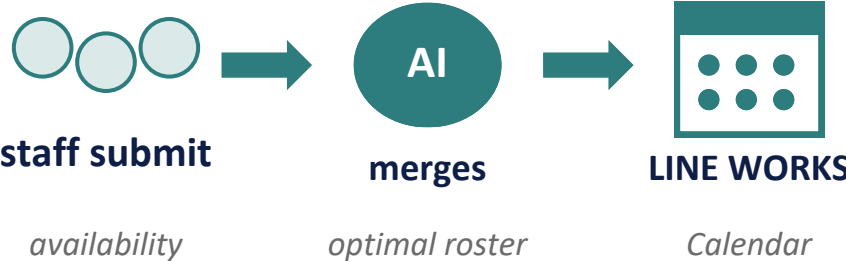
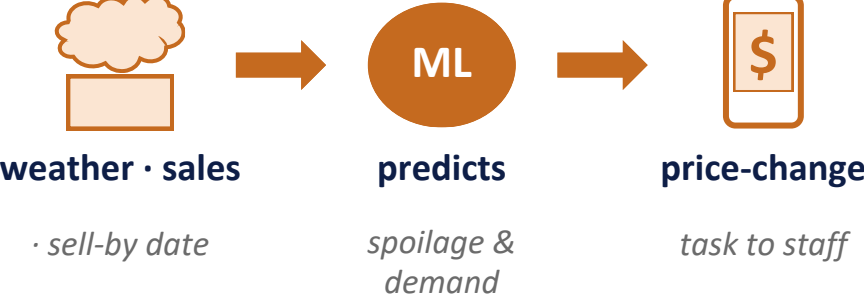


Four of five primary activities are deskless-heavy

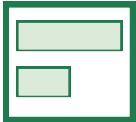
Five bottlenecks across the value chain

Operations	Operations	Marketing & Sales	Operations	Operations
1 Supplier & shift contact via personal LINE PAIN POINT High employee turnover means contact lists, supplier chats, and shift handovers all in employees' personal LINE accounts COST SIGNAL Security breach risk on every exit → Win on fresh food (supplier traceability)	2 Manual staff scheduling PAIN POINT Store manager spends time weekly merging staff availability into schedule COST SIGNAL Repetitive & time-consuming each week → Supermarket-for-everyone (operational excellence)	3 Demand & markdown downtime PAIN POINT Perishable pricing decisions made by gut; weather and event-driven demand spikes missed COST SIGNAL Shrinkage, lost sales, missed margin → Win on fresh food, Omnichannel integration	4 Repetitive inquiries disturb workflow PAIN POINT Staff repeatedly spend time on the same questions about SOP, salary, training, store policy COST SIGNAL Same questions interrupt floor workflow → Supermarket-for-everyone (operational excellence)	5 Manual shelf replenishment PAIN POINT Out-of-stocks discovered late by walking the aisle; expired items missed COST SIGNAL Late detection = lost sales + shrinkage → Supermarket-for-everyone (operational excellence)

AI candidates grouped by value chain stage (1 of 2)

STAGE	Operations	Operations	Marketing & Sales
PROBLEM	Supplier & shift contact via personal LINE <i>Contact info disappears when staff leave; no audit trail</i>	Manual staff scheduling <i>Manager spends time merging weekly availability</i>	Demand & markdown downtime <i>Perishable pricing decisions false; missed weather/event spikes</i>
HOW IT WORKS	 Bob <i>leaves</i> SEAT Assistant 1 <i>contacts stay</i> Andy <i>steps in</i>	 staff submit <i>availability</i> AI merges <i>optimal roster</i> LINE WORKS <i>Calendar</i>	 weather · sales <i>· sell-by date</i> ML predicts <i>spoilage & demand</i> price-change <i>task to staff</i>
AI SOLUTION	Supply Contact Management (#1) HQ-owned LINE WORKS seats by job role; preserve contact handover	Optimal Staff Scheduling (#2) AI merges staff availability → publishes schedule to LINE WORKS Calendar for everyone	AI Discount Pricing (#3) ML predicts spoilage and demand by location → pushes price-change tasks to floor staff via LINE WORKS

AI candidates grouped by value chain stage (2 of 2)

STAGE	Operations	Operations
PROBLEM	Repetitive Q&A from staff Floor staff repeatedly ask the same questions — SOP, salary, training, store policy	Manual shelf replenishment Out-of-stocks discovered too late by walking the aisle
HOW IT WORKS	 Staff asks SOP / HR / salary → BOT HyperCLOVA-X trained on guidelines →  Self-serve answer in LINE WORKS chat	 Smart shelf detects low stock →  Alert via LINE WORKS bot →  Replenish task pushed to floor staff
AI SOLUTION	PX Mart Internal Chatbot (#4) HyperCLOVA-X bot fed PX Mart guideline books and SOPs — every employee can self-serve in LINE WORKS chat	Sensor Shelf Monitoring (#5) Smart shelves detect product taken / about to run out → push replenish task to floor staff via bot

One example we exclude: real-time worker surveillance



AI-DRIVEN WORKFORCE SURVEILLANCE

Cameras, sensors, and wearables tracking employee speed, movement, and activity in real time

WHY WE'RE EXCLUDING IT

■ Measures the wrong thing

AI tracks physical motion — not work quality, customer-service value, or judgment under uncertainty.

■ Erodes trust irreversibly

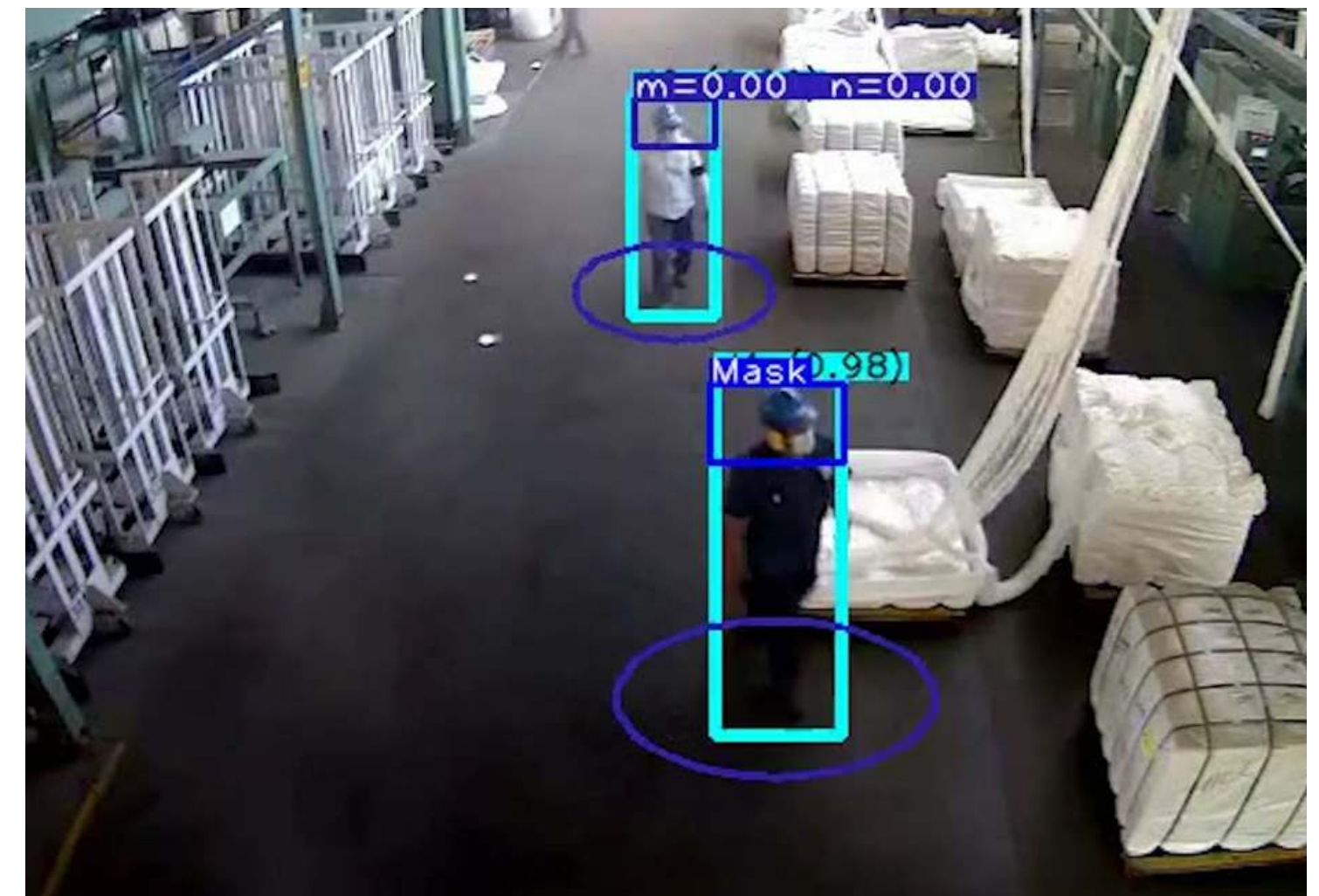
Once a workforce knows it's being surveilled, voluntary discretionary effort and upward problem-flagging collapse.

■ Asymmetric harm

Modest efficiency gains for HQ; significant psychological and dignity cost to a 30,000-person workforce.

■ Strategic backfire

PX Mart's brand is built on trust and neighborhood feel — surveillance signals the opposite at the customer-facing layer.

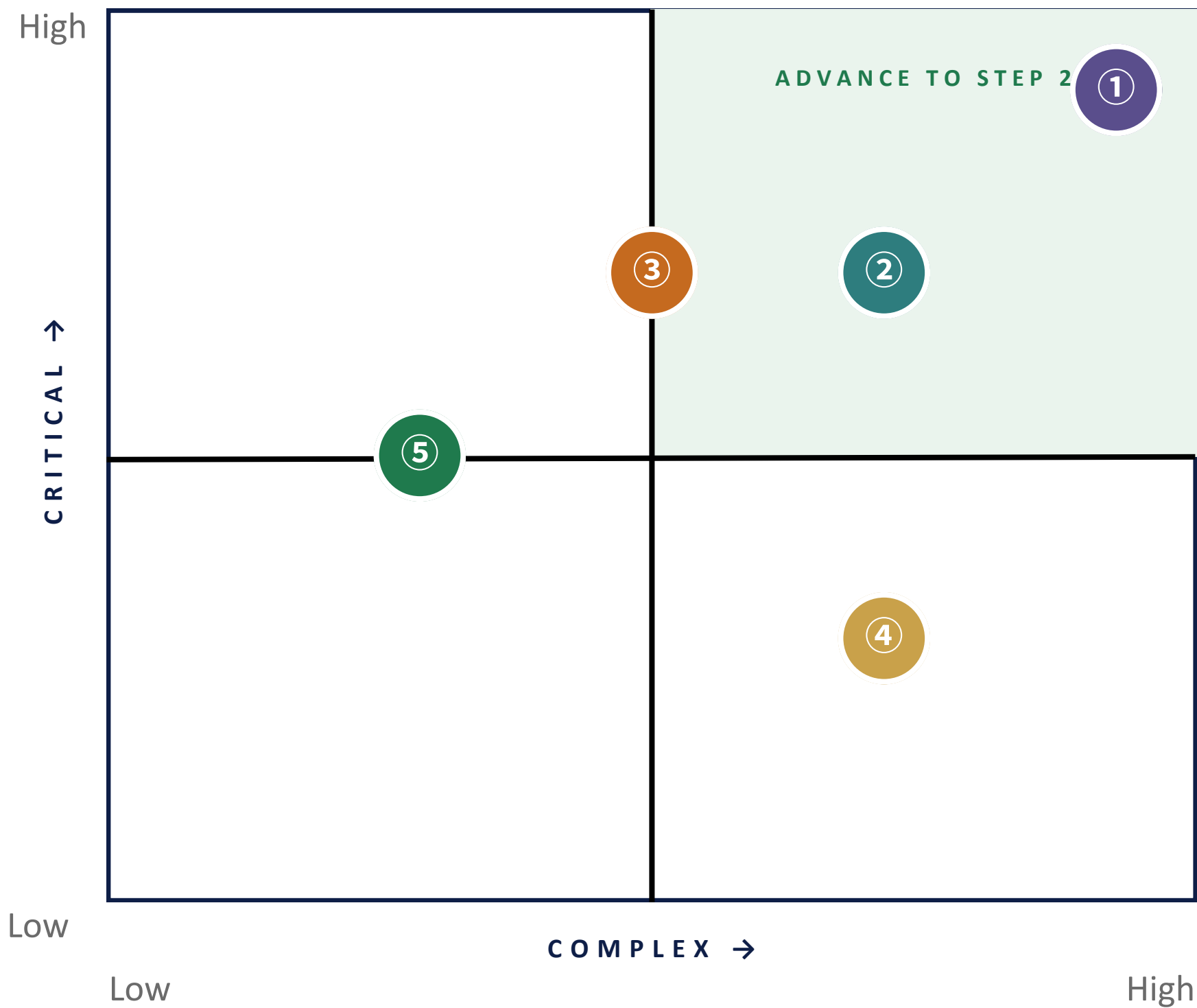


Five AI candidates — each with a strategic goal and a LINE WORKS implementation

Stage	No	Application	Pain point solved	LINE WORKS	Strategic goal supported
Operations	01	Supply Contact Management	Supplier & shift contact lives in personal LINE; lost on turnover	HQ-owned seats by role + admin console	<i>Win on fresh food (supplier traceability)</i>
Operations	02	Optimal Staff Scheduling	Manual roster building eats manager hours weekly	Calendar + survey native modules	<i>Supermarket-for-everyone (operational excellence)</i>
Marketing & Sales	03	AI Discount Pricing	Markdown decisions made by gut; demand spikes missed	Task push to specific employee	<i>Win on fresh food, Omnichannel integration</i>
Operations	04	PX Mart Internal Chatbot	Repetitive SOP/HR/training questions disturb workflow	Native HyperCLOVA-X bot in chat	<i>Supermarket-for-everyone (operational excellence)</i>
Operations	05	Sensor Shelf Monitoring	Out-of-stocks and expired items found too late	CV / sensor alerts pushed LINE WORKS task	<i>Supermarket-for-everyone (operational excellence)</i>

Step 1 — Critical × Complex filter

First filter from our whiteboard: only candidates that are BOTH critical and complex earn deeper scoring.



#	Application	Critical	Complex	Pass?
①	Supply Contact Management	✓	✓	ADVANCE
②	Staff Scheduling	✓	✓	ADVANCE
③	Discount Pricing	✓	✓	ADVANCE
④	PX Mart Chatbot	✗	✓	—
⑤	Sensor Shelf Monitoring	✗	✓	—

3 candidates advance to Step 2

Supply Contact Management · Staff Scheduling · Discount Pricing

PX Mart Chatbot and Sensor Shelf Monitoring are valuable but didn't pass the critical-AND-complex bar — revisit in Phase 2.

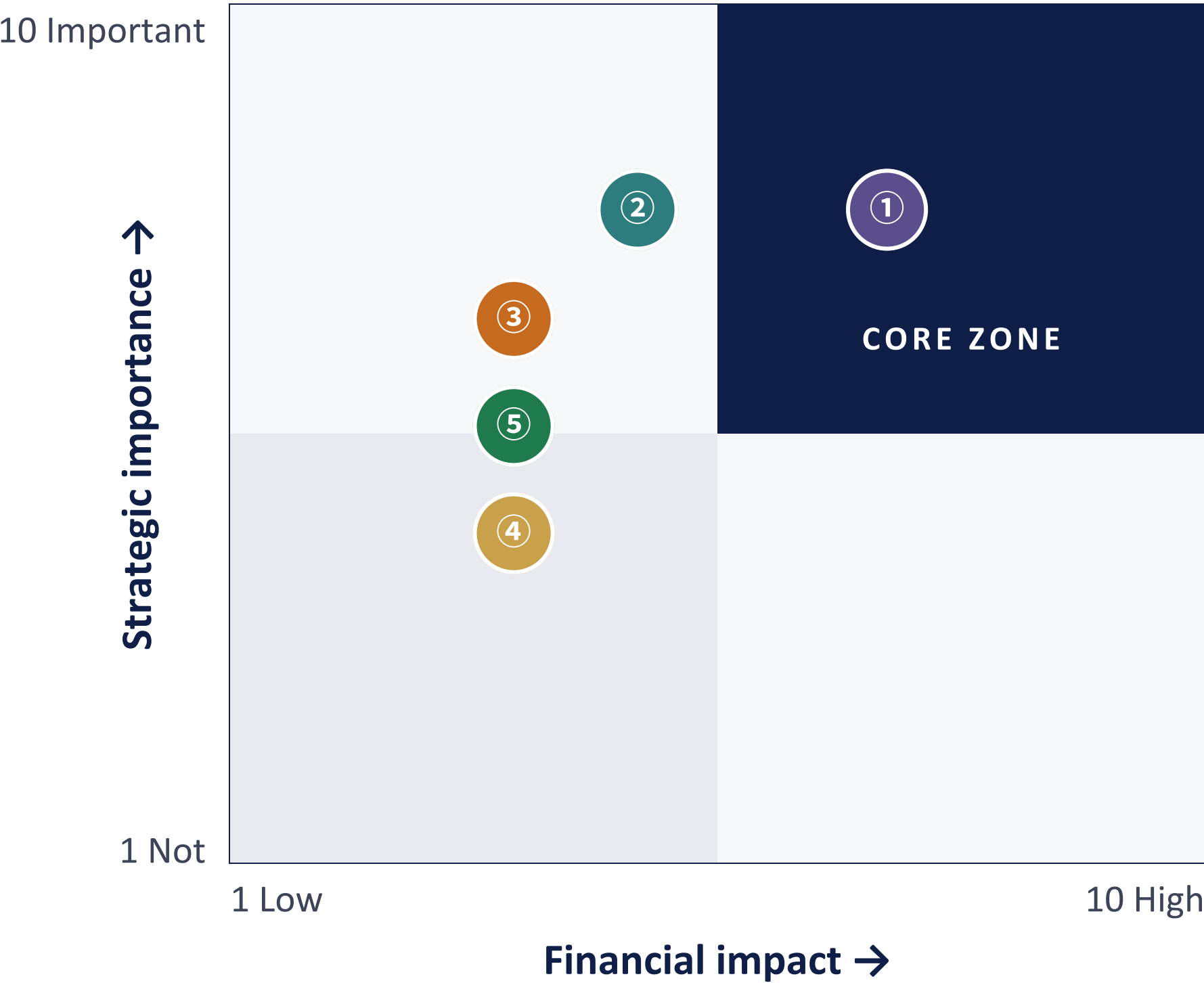
Core Area × Feasibility — visualizing the verdict

Higher core-area score = sits closer to PX Mart's strategic core. Higher feasibility = easier to build with LINE WORKS.



Strategic importance × financial impact (Prof. Kim methodology)

Scale: 1 = Very Important / High Impact, 5 = Average, 10 = Not Important / Low Impact



LEGEND

- ① Supply Contact Mgmt Core 8 · Feas 10
- ② Staff Scheduling Core 7 · Feas 9
- ③ Discount Pricing Core 7 · Feas 5
- ⑤ Sensor Shelf Core 7 · Feas 4
- ④ PX Mart Chatbot Core 5 · Feas 9

Verdict

Build #1: Supply Contact Management

Total 31 — wins on feasibility (admin console is native to LINE WORKS) and on core-area fit (closes the personal-LINE security gap left by part-time turnover).

Step 2 — Score survivors across four dimensions

Strategic Importance + Financial Impact + Core Area + Feasibility (each 1-10, higher = better).

#	Application	Strategic Importance	Financial Impact	Core Area	Feasibility	Total	Verdict
①	Supply Contact Management	8	5	8	10	31	BUILD #1
②	Staff Scheduling	6	5	7	9	27	BUILD #2
③	AI Discount Pricing	7	6	7	5	25	BUILD #3
⑤	Sensor Shelf Monitoring	8	5	7	4	24	Phase 2
④	PX Mart Chatbot	5	4	5	9	23	Phase 2

Highest total = 31

Supply Contact Management leads on every dimension. Wins by feasibility (LINE WORKS native admin module — almost zero build effort).

Strong runner-up = 27

Staff Scheduling: high feasibility (Calendar + survey), high core-area fit. Ready for parallel rollout.

Phase 1 closer = 25

Discount Pricing: high financial impact, but feasibility lower (needs an external ML model). Pilot in fresh-food category first.

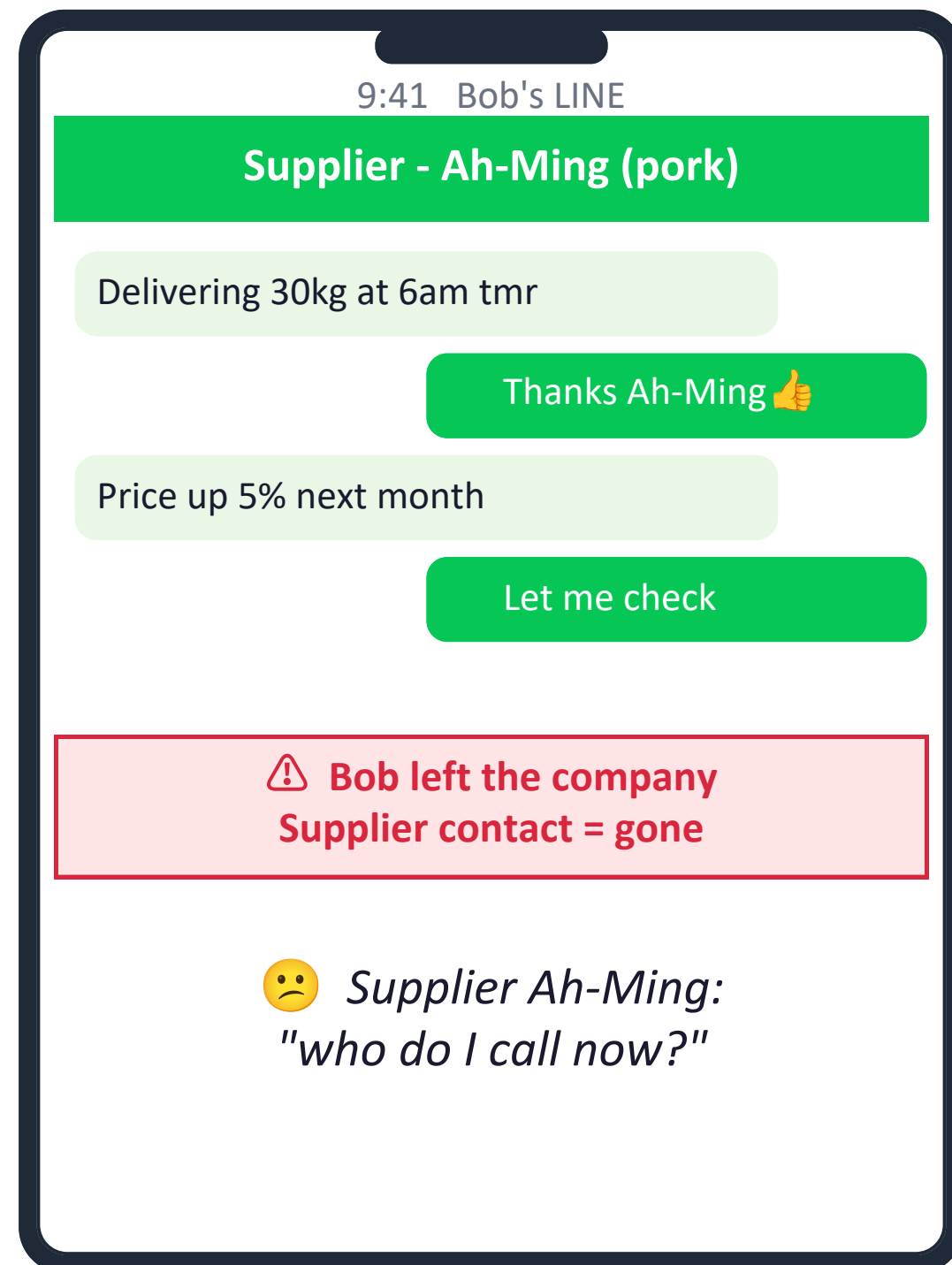
Supply Contact leads on feasibility; Staff Scheduling on time saved; Pricing on margin

01	Supply Contact Management FASTEST WIN · NATIVE BUILD HQ-owned LINE WORKS seats by job role. When a worker leaves, the seat (and its supplier contacts, vendor chats, supplier handover history) stays. New person — say Andy — is dropped into Bob's seat, customers and suppliers see no disruption.	KEY TRADE-OFF <i>Requires HQ to redesign job-to-seat mapping for all 1,200+ stores. Change-management effort, not technical effort.</i>
02	Optimal Staff Scheduling BIGGEST TIME SAVED Each staff submits their availability via LINE WORKS survey. AI merges constraints into the optimal weekly roster, then publishes to LINE WORKS Calendar — every employee and store manager sees the same schedule.	KEY TRADE-OFF <i>Lower direct P&L impact than #3; benefit shows up in manager hours freed, not gross margin.</i>
03	AI Discount Pricing BIGGEST MARGIN WIN An external ML model (location, weather, sell-by date, foot traffic) predicts when fresh items will spoil and what markdown drives clearance. The pricing recommendation pushes as a LINE WORKS task to floor staff: change the price, swap the item.	KEY TRADE-OFF <i>Multi-quarter to validate the ML model; pilot in fresh-food category in 50 stores before fleet-wide.</i>

SOLUTION #1

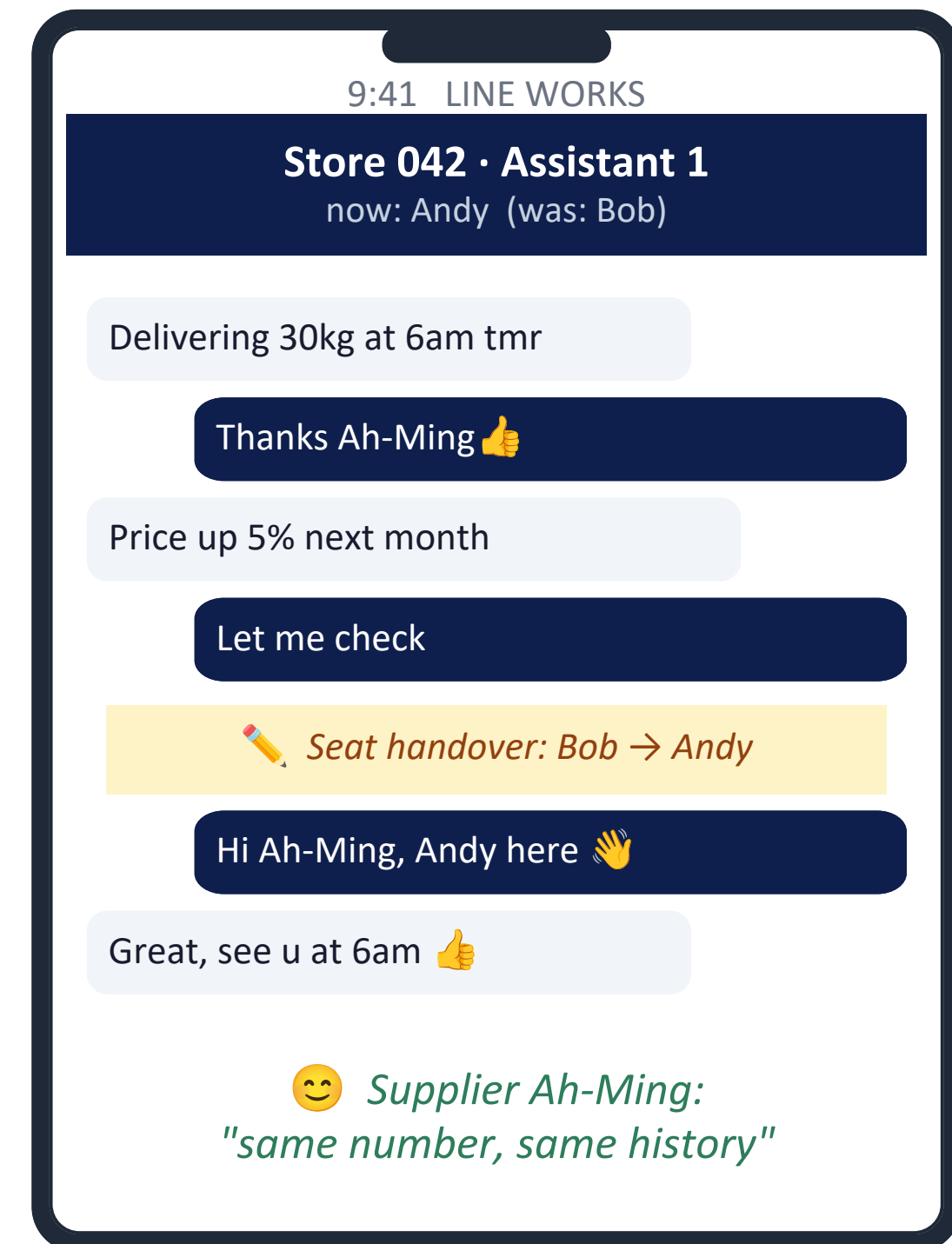
Supply Contact Management, what a supplier actually sees

TODAY · personal LINE



➡
LINE WORKS
rebuilds it

WITH LINE WORKS · HQ-owned seat



The supplier chats with a role, not a person. Turnover stops being a security problem

Optimal Staff Scheduling, what the manager encounter

TODAY · manual scheduling

WITH LINE WORKS · AI Calendar published

Week of Jun 10 — Store 042

Mon

AM:

Mei, Joe, ~~X~~Lin sick

Mon

PM:

??? need cover

Tue

AM:

Joe, Sam, Mei

Tue

PM:

Lin? — texted no reply

Wed

AM:

Sam, Mei + 1?

Thu

...

STILL WORKING ON IT

Fri

...

XXX rewriting

Sat

...

???

3 hrs and still not done!

LINE WORKS rebuilds it

July 17

LINE WORKS Calendar · Store 042 · Week of Jun 10

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
AM	Mei	Joe	Sam	Mei	Joe	Sam	Mei
PM	Joe	Lin	Mei	Sam	Lin	Mei	Joe
EVE	Sam	Mei	Joe	Lin	Sam	Joe	Lin

✓ AI-generated · 12 sec

SOLUTION #3

AI Discount Pricing, what the floor staff had to do

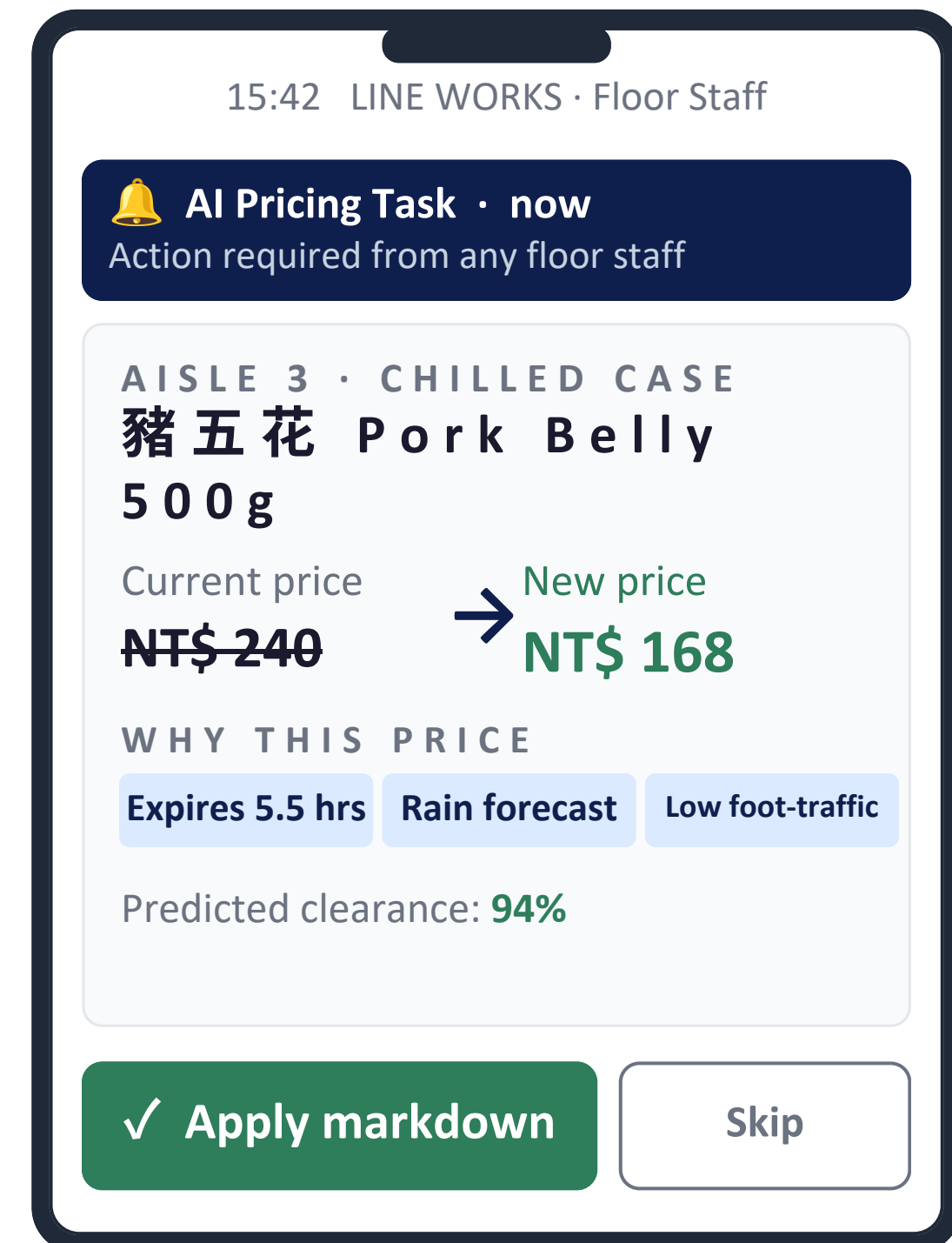
TODAY · unreliable markdown sticker



🤔 **Store manager:**
"50% off? 30%? I dunno. Just slap it on."
→ leaves margin on the floor OR shrinkage

➡
*LINE WORKS
rebuilds it*

WITH LINE WORKS · AI task pushed to floor staff



ML predicts spoilage & demand by location → AI sends a one-tap price-change task to the floor.

Begin with Supply Contact Management

RECOMMENDATION

Supply Contact Management

HQ-owned LINE WORKS seats by job role. Suppliers and shift partners contact a SEAT, not a person. Turnover stops being a security event.

WHY THIS, FIRST

- Highest total score (31/40) on our four-dimension matrix
- Closes a real security gap: 30%+ part-time turnover means contact lists walk out the door each year
- Built natively in LINE WORKS — admin console + permissions, no new hardware, no new vendor
- Bob (assistant role) leaves → seat stays, contacts intact → Andy steps in seamlessly
- Quick win earns organizational permission for harder AI projects later

STRONGEST COUNTERARGUMENT

"Reorganizing 30,000 employees onto seat-based identities is just bureaucratic reshuffling — it doesn't generate revenue."

OUR RESPONSE

Phased rollout

Pilot in 1 supplier-heavy distribution centre (~50 staff) for 3 months. Validate handover quality, then expand store-by-store.

Already paid for

LINE WORKS admin console comes bundled — no incremental software cost. ROI is in risk reduction, not new revenue.

Foundation play

Once seats and roles are clean in LINE WORKS, every later AI project (scheduling, pricing, chatbot) inherits the same identity layer for free.

Automate the handover, but augment the human relationship

AUTOMATE

THE SEAT & THE HANDOVER

What the system does on its own:

- Automatically provision and revoke LINE WORKS accounts based on HR events
- Transfer supplier communication history and ownership during role changes
- Auto update the role title visible to suppliers (“Andy - Assistant 1” instead of “Bob”)
- Enforce centralized access and permission policies
- Audit log every supplier conversation for compliance

AUGMENT

THE HUMAN RELATIONSHIP

What the human still owns:

- First conversation with a new supplier (building trust)
- Negotiation on price, quality, and delivery timing
- Operational judgment during supplier disruptions (late, damaged)
- Welcoming the new seat holder and introducing them to suppliers personally
- Local knowledge: which supplier is reliable in typhoon season, which is flexible on volume

While automation improves efficiency, it is equally important to use augmentation to preserve human relationships and decision-making.

Conclusion

1

MECE pyramid, top-down, bottom up

We started with LINE WORKS' four fit conditions -> PX Mart strategic goals -> the value chain -> five MECE-grouped AI applications. Each layer narrows mutually exclusively.

2

Two-step methodology

Step 1 (Critical × Complex) filtered 5 candidates to 3. Step 2 (Strategic + Financial + Core Area + Feasibility) scored survivors. Highest score: Supply Contact Management at 31/40.

3

First project: Supply Contact Management

Native LINE WORKS build, zero new hardware, closes the personal-LINE security gap left by 30%+ part-time turnover, creates the identity layer every later AI project will inherit.

4

Phase 2 roadmap is already mapped

Staff Scheduling and Discount Pricing follow next. Sensor Shelf Monitoring and the internal Chatbot move to a later phase — not because they're weak, but because they didn't pass the critical-AND-complex filter.

Thank you